RW-6110 **501101**

M.Sc., DEGREE EXAMINATION, NOVEMBER 2010

Bio-Technology

BIOCHEMISTRY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Glycerophospholipids

2. Structure of glycogen

3. Simple diffusion

4. ETC inhibitors

5. Structure of ATP

6. Redox reactions

7. HMP shunt

8. Ribozyme

9. Ammonium sulfate fractionation

10. Energetics of TCA cycle

Part - B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Role of buffers in biological system.

(*Or*)

(b) Explain the classification of amino acids.

12. (a) Explain active and passive transport with example.

(*Or*)

(b) Comment on oxidative phosphorylation.

13. (a) Briefly explain the Second Law of Thermodynamics.

(*Or*)

(b) Role of ATP synthase in mitochondrial potential ?

14. (a) Explain purine biosynthesis pathway.

(*Or*)

(b) How is citric acid cycle regulated.

15. (a) Comment on the industrial applications of enzymes.

(*Or*)

(b) Derive a relationship between substrate concentration and initial velocity.

Part - C (3 × 10 = 30)

Answer any **three** questions.

16. What are co-enzymes ? Explain its significance.

17. Explain fluid mosaic model of plasma membrane ?

18. Write notes on the following:

(a) Classification of enzymes (2)

(b) First law of Thermodynamics (2)

(c) Uncouplers (2)

(d) Antimycin (2)

(e) Gibbs free energy (2)

19. Explain the steps involved in glycolysis ?

20. Explain the steps involved in the purification of enzymes.

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RW-6111 | 501102

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

MICROBIOLOGY

(CBCS–2008 onwards)

Time : 3 Hours | Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

Write notes on :

1. Elie Metchnikoff.

2. Five kingdom classification.

3. Inclusion bodies.

4. Prions.

5. Oncogenic virus.

6. Diptherial toxin.

7. Merits and demerits of culture dependent method.

8. Isotopic probing.

9. Camemberti cheese.

10. Xenobiotics.

Part - B (5 × 5 = 25)

Answer **all** questions.

All questions carry equal marks.

11. (a) Write a note on basic criteria for bacterial classification.

(*Or*)

(b) Discuss on modern developments in microbiology.

12. (a) Discuss in detail about flagella and its types.

(*Or*)

(b) Write about general properties of viruses.

13. (a) Write the etiology, symptoms and control of UTI.

(*Or*)

(b) Describe about Kirby Bauer method of sensitivity tests.

14. (a) What is DGGE ? Explain the method and its application.

(*Or*)

(b) Explain the process of sequence driven analysis.

15. (a) Enumerate the role of microorganisms in soil.

(*Or*)

(b) Explain the process of beer production.

Part - C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Describe briefly the salient features of different classes of bacteria.

17. Explain prokaryotic cell structure and organization.

18. Explain the pathogenicity of :

(a) *Staphylococcus.*

(b) Rabies virus.

19. What are metagenomics ? Explain the construction of metagenomic library.

20. Explain in detail about mechanism on biodegradation of xenobiotics.

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RW-6112 **501103**

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

MOLECULAR BIOLOGY AND GENETICS

(CBCS–2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Define Wobble hypothesis.

2. What is the importance of Gierer and Schramm Tobacco Mosaic Virus (TMV) experiment ?

3. What is a dilysogen ?

4. What is Ter cutting rule ?

5. Comment on the phenotype of *pro*-35 (Am).

6. Hfr.

7. Features of Tn10.

8. What is the difference between generalized and specialized transduction ?

9. VNTR.

10. Name two oncogenes.

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) Describe the contributions of Avery et al.

(*Or*)

(b) Describe the Post Translational modification of proteins.

12. (a) Describe the molecular events associated with lysogenic cycle of Lambda phage.

(*Or*)

(b) Write notes on the positive regulation of lac operon.

13. (a) Compare and contrast a genetic selection from a genetic screen.

(*Or*)

(b) Describe genetic mapping using two factor and three factor crosses.

14. (a) Describe replicative transposons with example.

(*Or*)

(b) Describe the molecular events associated with specialized transduction.

15. (a) Describe the structural organization of human chromosome.

(*Or*)

(b) Mention the outcome of Human Genome Project.

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. Describe the semi conservative DNA replication in prokaryote.

17. Describe the role of SOS repair in mutagenesis.

18. Describe an operon you have studied which is involved in a biosynthetic pathway.

19. Describe the molecular mechanism of cancer.

20. Describe the various uses of mutants.

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RW-6113 **501104**

M.Sc., DEGREE EXAMINATION NOVEMBER 2010

Biotechnology

CELL BIOLOGY

(CBCS–2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Cell fusion.

2. Cyclin-Dependent kinase.

3. Microtubule-Associated protein.

4. Structure of Neuron.

5. Principle of electron microscopy.

6. Structure of mitochondria.

7. Polyprotein genes.

8. Avert, Mccleod and Mccarthy's experiment.

9. What is photorespiration ?

10. How Retrovirus can act as vector for Oncogene.

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain in detail about the active transport by ATP-powered pumps.

(*Or*)

(b) Describe the various phases of meiosis.

12. (a) Give the structural illustration of RBC membrane model.

(*Or*)

(b) Describe briefly the structure of the food conducting tissue in plants.

13. (a) Why do multicellular organisms able to grow much larger than unicellular organisms ?

(*Or*)

(b) Enumerate the most important function of fatty acids in a living cell.

14. (a) Sperm is considered as motile cell, justify and give reasons with suitable sketch.

(*Or*)

(b) Explain the significance of crossing over.

15. (a) Describe the structural and functional properties of cellular enzyme.

(*Or*)

(b) Give an account on the regulators for cytoplasmic-nucleus interaction.

Part - C (3 × 10 = 30)

Answer any **three** questions.

16. What advantages does an eukaryotic cell gain by nuclear envelope ? Why does nuclear envelope have pores, while other membrane of the cell do not ?

17. Discuss about the protein trafficking from endoplasmic reticulum to golgi.

18. Explain about the nature, composition and organization and function of plant cell wall.

19. Describe the cycle of events occurring after binding a ligand to receptor Tyrosine kinase and activation of Ras.

20. Describe about the theories regarding tumor formation and Aging.

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RW-6114 | 501105

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

LAB I—ANALYTICAL BIOCHEMISTRY

(CBCS–2008 onwards)

Time : 3 Hours | Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Define Beer-lamberts law.

2. Define Molar Extinction coefficient.

3. Differentiate between single beam and double beam spectrophotometers.

4. Significance of enzymes in diagnosing liver damage.

5. Applications of radioisotopes in medicine.

6. Define the units Curie and Becquerel.

7. Role of TEMED in PAGE.

8. Lipase and pancreatitis.

9. Principle of Bradford's method for protein estimation.

10. Define Rf value.

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) Elaborate on determination of fasting, post-prandial and random blood sugar levels.

(*Or*)

(b) Explain the principle and methodology for estimation of carbohydrate by DNSA method.

12. (a) Discuss the estimation of serum cholesterol and evaluation of risk of coronary heart disease.

(*Or*)

(b) Write in detail on spectrophotometry and its applications.

13. (a) Explain the methodology for the determination of pKa value of acetic acid.

(*Or*)

(b) Discuss the Henderson-Hasselbach equation and its applications.

14. (a) If two proteins have the some molecular weight, but different charge properties, which technique is employed to separate them ? Explain.

(*Or*)

(b) Explain the principle of measurement of pH using pH meter.

15. (a) Write in detail about identification of spots in paper chromatography.

(*Or*)

(b) Differentiate between Ultrafiltration and Dialysis.

Part - C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the separation of compounds by two way Paper chromatography.

17. Discuss the separation of plant pigments by TLC.

18. Write in detail on separation of proteins by size exclusion chromatography.

19. Enumerate on the applications of PAGE in separation of nucleic acids.

20. (a) Derive the Michaelis-Menten equation.

(5 marks)

(b) How will you determine the Vmax and Km values of enzyme catalyzed reactions by double-reciprocal plot ?

(5 marks)

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RW-6115 | 501106

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

LAB II—MICROBIOLOGY

(CBCS–2008 onwards)

Time : 3 Hours | Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Pasteurization.

2. Ascospores.

3. Distinguish between Fastidious and Non-fastidious medium.

4. Lactophenol cotton blue.

5. Explain the role of Amylase.

6. Pheramones.

7. How will you differentiate *E. coli* from *E. aerogenes* ?

8. How the bacterial organism degrades H_2O_2 ?

9. Pour plate technique—Explain.

10. What is McFarland standard ?

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain in detail about the enumeration of fungi from soil.

(*Or*)

(b) How will you purify bacterial culture ? Explain.

12. (a) Discuss in detail about the impact of environmental factors on bacterial growth.

(*Or*)

(b) Write in detail about the slide culture technique and its applications.

13. (a) Define media and give short notes on role of selective media in bacterial cultivation.

(*Or*)

(b) Mention about the composition of nutrient agar medium and explain the role of each components.

14. (a) Write in detail about the cultivation of anaerobic bacteria.

(*Or*)

(b) What are probiotics ? Explain in detail about its mode of actions.

15. (a) Explain in detail about IMVIC test and its significance.

(*Or*)

(b) Give a detail account on Kirby-Bauer test.

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about the various types of sterilization and disinfectant techniques.

17. Write in detail about the principle and procedures for the following :

    (a) Negative staining.

    (b) Differential staining.

18. Explain in detail about the screening of antibiotic producing organisms from environmental sample.

19. How will you analyze the potability of water through Membrane Filtration technique and narrate its advantages ?

20. Discuss in detail about bacterial cell to cell communication systems and its role in various physiological activities of bacterial community.

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RW-6116 | 501501

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

Elective I—BIOPHYSICS AND INSTRUMENTATION

(CBCS–2008 onwards)

Time : 3 Hours | Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Define Peptide bond.

2. What is hydrophilic interaction ?

3. Discuss about secondary structure of protein.

4. Define DNA polymorphism.

5. What are called as radio-isotopes ?

6. What is meant by scintillation counter ?

7. Define Affinity chromatography.

8. Explain Mobile Phase and Stationary Phase.

9. Describe briefly about how the X-Ray diffraction pattern is produced.

10. Define Fluorescence microscopy.

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) Describe the strong interactions in macromolecules.

(*Or*)

(b) Discuss about the structure and properties of water.

12. (a) Write the importance of Ramachandran plot in protein structure analysis.

(*Or*)

(b) Describe about the various structural polymorphism of DNA and RNA.

13. (a) Discuss about the measurement of radioactivity in biological sample.

(*Or*)

(b) Explain the devices used to measure radiations.

14. (a) Define Chromatography. Write about the principle of Chromatography.

(*Or*)

(b) What are the principles and application of Agarose gel electrophoresis ?

15. (a) What is NMR and how it is used in biological research ?

(*Or*)

(b) Describe the basic principles of Transmission and Scanning electron microscopy.

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. What is meant by interactions ? Explain the different interactions used to stabilize the macromolecular structures.

17. Describe about different level of protein structure organization. Describe about protein folding.

18. What are the applications of radioactive isotopes in biological samples ?

19. Describe the principles of sedimentation. Discuss about the various types of centrifuges and rotors used for it.

20. Write about the principles of biophysical methods used for analysis of biopolymer structures. Discuss in detail with X-Ray and ESR spectroscopy.

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RW-6117 | 501301

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

BIOINFORMATICS

(CBCS–2008 onwards)

Time : 3 Hours | Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. What are the secondary database available in web ?

2. Define Open source software's.

3. Write the use of local alignment.

4. What is known by database search ?

5. Write the name of the software's used for multiple sequence alignment.

6. Write the uses of phylogenetic analysis.

7. What is called as motif ?

8. Why analysis of Protein secondary structure is important ?

9. What are the methods available for DNA sequencing ?

10. What is meant by primer design ?

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) Discuss about retrieving information and sequence from databases.

(*Or*)

(b) Write any five important commands used in RASMOL.

12. (a) Describe the Dynamic programming algorithms.

(*Or*)

(b) Write three principles of sequence similarity search algorithms.

13. (a) Discuss about format and editing in multiple sequence alignments.

(*Or*)

(b) Discuss about multiple sequence alignment in CLUSTAL W.

14. (a) Discuss any two available methods of protein secondary structure prediction.

(*Or*)

(b) Briefly discuss about Drug designing concepts.

15. (a) What is meant by sequence assembly and finishing in DNA sequencing ?

(*Or*)

(b) Discuss about Computing in Proteomics.

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. Explain any ten important commands in EMBOSS.

17. Describe the different basic local alignment tools available in the web. Explain FASTA sequence format file.

18. Describe the progressive methods of local Multiple Sequence Alignment with two available programs.

19. Discuss about any one of tertiary structure prediction methods available in web with example.

20. Explain about prediction of Genes and Regulatory sequence in DNA.

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RW-6118

501302

M.Sc., DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

ANIMAL BIOTECHNOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Define cell lines.

2. Embryonic stem cell.

3. Vaccinia virus.

4. Recombinant animal viral vectors.

5. Regulatory proteins.

6. Neurodegenerative disorders.

7. Gene therapy for HIV diagnosis.

8. Synthetic viral vectors.

9. Stem cell differentiation.

10. Liposome mediated gene transfer.

Part - B (5 × 5 = 25)

Answer **all** questions.

11 (a) Explain various gene transfer methods.

(*Or*)

(b) Explain Reproduction Biotechnology techniques.

12. (a) What are RNA viral vector ? Explain.

(*Or*)

(b) Explain the construction of recombinant animal viral vectors.

13 (a) Application of yeast system to study eukaryotic gene function.

(*Or*)

(b) Role of animal biotechnology in production of recombinant vaccines.

14 (a) Biotechnology applications for HIV diagnosis.

(*Or*)

(b) Gene therapy – *Ex vivo*.

15 (a) What is meant by stem cell differentiation ?

(*Or*)

(b) Explain preparation and application of umbilical cord blood cells.

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. What are transgenic animals ? Explain.

17. Explain the different types of animal viral vectors.

18. Explain the applications of animal biotechnology in various fields.

19. Explain DNA based diagnosis of genic disease.

20. Bioethics and stem cell research.

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RW-6119

501303

M.Sc., DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

MARINE BIOTECHNOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Agar agar.

2. Ecteinascidin

3. Hydrothermal vents.

4. Barophils.

5. Western blotting.

6. Vibriosis.

7. Polyploidy.

8. Transposon.

9. Avirulence.

10. QS inhibitors.

Part - B (5 × 5 = 25)

Answer **all** questions.

11 (a) What is phyco-remediation ? Explain briefly.

(*Or*)

(b) Marine sponges are potential source of drugs—Justify.

12. (a) Write briefly about extremophils.

(*Or*)

(b) Write a short note on Hyperthermophils.

13 (a) Discuss shortly about the Southern blotting technique.

(*Or*)

(b) Write a brief note on the viral pathogens of shrimps.

14 (a) Explain briefly about transgenic fish with an example.

(*Or*)

(b) Write a brief note on sex reversal in aquaculture species.

15 (a) Explain shortly about the types of autoinducers.

(*Or*)

(b) Discuss the role of QS inhibitors in expression of violence gene.

Part - C (3 × 10 = 30)

Answer any **three** questions.

16. Write a critical essay on marine resources of India.

17. Explain in detail about the steps involved in the culture of unculturable bacteria.

18. Describe the application of PCR in aquaculture industry.

19. Explain Androgenesis, Gynogenesis and hybridization with suitable flow diagrams.

20. Give detailed account of Quorum sensing and its significance.

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RW-6120

501304

M.Sc., DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

LAB V : RECOMBINANT DNA TECHNOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. What is the reason for getting genomic DNA contamination while isolating plasmid DNA ?

2. What is the role of DEPC in RNA isolation ?

3. Comment on Phagemic with an example.

4. Comment on hybridization probe.

5. What is the difference between DNA Taq polymerase, HotStart Tag polymerase and pfu Tag polymerase enzymes ?

6. What is the role of $MgCl_2$ in Polymerase Chain Reaction ?

7. How will you prepare Polyacrylamide gel for SDS-PAGE ?

8. What are all the basic features for an expression vector ?

9. Comment on DNase I foot printing.

10. What are all the common challenges of DNA sequencing ?

Part - B (5 × 5 = 25)

Answer **all** questions.

11 (a) Describe the principle and procedure involved in the isolation of Plasmid DNA by Boiling Lysis method. Compare this method with alkaline lysis method.

(*Or*)

(b) Describe the quality and quantity checking of the Nucleic acids by UV spectrophotometer and AGE ?

12. (a) Explain any four cloning vectors and their applications.

(*Or*)

(b) Explain the principle and procedure of Southern blotting.

13 (a) Comment on the following :

(i) Short tandem repeats.

(ii) Amplified fragment length polymorphism.

(*Or*)

(b) Explain the principle and detailed procedure for molecular identification of bacteria by PCR.

14 (a) Explain the stages in Downstream Processing.

(*Or*)

(b) Explain the process in real time and reverse transcriptase PCR.

15 (a) Give an account on transcript mapping.

(*Or*)

(b) Explain the effective method of site directed mutagenesis.

Part - C (3 × 10 = 30)

Answer any **three** questions.

16. What are all the precautionary steps involved in RNA isolation and the method of isolation of Eukaryotic RNA using TRIZOL ?

17. Write the steps involved in the construction of genomic DNA library.

18. Explain in detail about multilocus and singlelocus DNA profiling.

19. Explain the analysis of proteins expressed by SDS-PAGE and Western blotting.

20. Describe the method of automated DNA sequencing and its application.

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RW-6121

501305

M.Sc., DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

LAB I : PLANT BIOTECHNOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.
All questions carry equal marks.
Provide illustration wherever necessary.

1. Minor salts.

2. Cotyledonary node.

3. Multiple shoots

4. Satellite markers.

5. Kinetin.

6. Polyamines.

7. T-DNA.

8. *Agrobacterium rhizogenes*.

9. Transformants.

10. x-glu.

Part - B (5 × 5 = 25)

Answer **all** questions.
All questions carry equal marks.

11 (a) Write about Half strength MS medium.

(*Or*)

(b) Explain sterilization—Describe.

12. (a) Explain In vitro seed germination.

(*Or*)

(b) What is somatic embryogenesis ?

13 (a) What is synseed ? How it is prepared ?

(*Or*)

(b) Hairy root culture – Explain.

14 (a) Extraction and purification of chromosomal DNA from plants.

(*Or*)

(b) Give details on role of reporter genes in transformation.

15 (a) Role of Ti plasmid in gene transfer.

(*Or*)

(b) What is GUS histochemical staining ? Explain.

Part - C (3 × 10 = 30)

Answer any **three** questions.
All questions carry equal marks.

16. Give a detailed account on Sterilization in Plant Tissue culture.

17. Describe about Indirect Organogenesis.

18. Write a detailed notes on Plant Growth Regulators.

19. What is vector and how it is constructed ?

20. Give an account on expression of gene in Transformed plants.

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RW-6122

501510

M.Sc., DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

Elective IV : IPR, BIOSAFETY AND BIOETHICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Differentiate between a patent and know-how.

2. Brief on WTO.

3. Write a note on GATT.

4. Differentiate between genetic testing and genetic screening.

5. What do you mean by biodiversity ?

6. What are ELSI associated with it ?

7. State the different forms of intellectual protection.

8. What is sustainable development ?

9. What is Biological contaminants ?

10. What is copy right ?

Part - B (5 × 5 = 25)

Answer **all** questions.

11 (a) What are the ELSI arising out of the Human Genome Project.

(*Or*)

(b) Comment on the Licensing of Flavr Savr ™ tomato.

12. (a) What is the difference between Trademark and copyright.

(*Or*)

(b) What are the guidelines for doing research in transgenic plants.

13 (a) What are the different steps in filling patents ?

(*Or*)

(b) Why preparing of scientific developments in term of product and service has become necessary in current scenario.

14 (a) Elucidate the product development process from idea conception to beyond.

(*Or*)

(b) Discuss in detail about types of Intellectual Property Rights.

15 (a) What is the role of patents in biotechnological development ?

(*Or*)

(b) Describe the functions of GATT, WTO and WIPO.

Part - C (3 × 10 = 30)

Answer any **three** questions.

16. Outline the process in the filing of a patent. What is the post-acceptance process ? Mention the salient features of the right of priority of the Parts convention. What is meant by opposition and infringement of a patent.

17. What are the ethical concerns associated with recombinant DNA technology and human cloning. Formulate ethical guidelines that should govern biotechnology research in these areas.

18. What are the ELSI in human animal experimentation ? Should animal experimentation be banned. Voice your opinions from an ethical stand point. Outline the salient features of the Nurember code and the Helsinkig protocol which guide researchers involved in human experimentation.

19. Discuss classification of organisms based on risks for laboratory workers and the environment. Outline the risk groups and biosafety level. Discuss spill protocols and waste disposal precautions and methods, regulatory compliance and containment.

20. What is biodiversity ? How can the nation's biodiversity be protected ? What measure should be taken to achieve sustainable development. What are the implications of WTO on biodiversity.

——— *** ———

RW-6123

501512

M.Sc., DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

Elective V : FUNCTIONAL GENOMICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Define Probe in DNA microarray and Northern Blotting.

2. What is SAGE ?

3. Explain the steps involved in 2-DGE briefly.

4. Explain in brief about DD-PCR.

5. What is Mass spectrometry ?

6. Uses of reporter gene(s) in Proteomics.

7. Comment on host-pathogen interaction using *C. elegans*.

8. Comment on the uses of mice for functional genomic studies.

9. Define pharmacogenetics.

10. What are the main challenges in drug discovery process.

Part - B (5 × 5 = 25)

Answer **all** questions.

11 (a) Describe the chemistry involved in SYBR green and Taqman probes.

(*Or*)

(b) Explain in detail about Standard Subtractive Hybridizaiton (SSH).

12. (a) Advantages and disadvantages of 2D gel electrophoresis for Protein analysis.

(*Or*)

(b) Give short notes on different Ionization sources of Mass Spectrometry.

13 (a) Advantages and disadvantages of yeast two-hybrid system.

(*Or*)

(b) How to characterize a protein for its functionality ?

14 (a) Raitonale for using *C.elegans* over other model systems for functional genomic analysis.

(*Or*)

(b) Comment on the role of pathogenicity islands of microbial pathogens.

15 (a) Methods to identify the potential drug targets.

(*Or*)

(b) Explain in detail about personalized medicine.

Part - C (3 × 10 = 30)

Answer any **three** questions.

16. Write in detail about role of Microarray in Medicine.

17. Principle and working Procedure of 2D NMR.

18. Advantages of tagging GFP for localization studies.

19. Role of any two environmentally significant microorganisms.

20. What are the high throughout technologies available for screening a drug ?

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RW-6124

501201

M.Sc., DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

IMMUNOBIOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Comment on Thymus.

2. Differentiate active immunity from passive immunity.

3. Immunoglobin is formed in response to an antigen when it enters into the body—Justify.

4. What are the significances of haptens ?

5. What are the various functions of complements ?

6. Distinguish between cytosolic and endocytic pathways.

7. Comment on HLA.

8. Comment on macrophages.

9. Define Allergy.

10. How immunological methods are useful for the production of vaccines.

Part - B (5 × 5 = 25)

Answer **all** questions.

11 (a) Describe the various primary lymphoid organs.

(*Or*)

(b) How cell mediated immunity is achieved ?

12. (a) Explain the general structure of immunoglobin.

(*Or*)

(b) Explain the salient features and functions of cytokines.

13 (a) Discuss the biological effects of complements.

(*Or*)

(b) Discuss the mechanism of endocytic pathway.

14 (a) Discuss the structure of class I MHC and class II MHC.

(*Or*)

(b) Discuss the mechanism of immunity to bacterial and viral infection.

15 (a) Discuss about Oncogenes.

(*Or*)

(b) Analyse the various immunotherapy for cancer treatment.

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. Explain the types and mode of action of innate immunity.

17. Discuss the structure and functions of IgA and IgM.

18. Analyse the mechanism of different types of passive immunization. Add a note on comparison of active and passive immunization.

19. Describe the organ specific and non-organ specific autoimmune disorders.

20. Discuss about various disorder due to immunodeficiency.

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RW-6125

501203

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

PLANT MOLECULAR BIOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Transplastomic plants.

2. cp Proteins.

3. RNA Termination.

4. cDNA.

5. Germplasm Preservation.

6. Agro infection.

7. Ri Plasmids.

8. Opines.

9. Proteobacteria.

10. Cytochrome b.

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) Give an account on Chloroplast genome.

(*Or*)

(b) Write notes on :

(i) Promiscuous DNA.

(ii) Mitochondrial replication.

12. (a) Describe the principle and application of artificial seed in commercial seed industry.

(*Or*)

(b) Give an account on molecular markers.

13. (a) Explain Bacterial conjugation.

(*Or*)

(b) Describe Ti plasmid.

14. (a) Explain PHB.

(*Or*)

(b) Explain about artificial mode of gene transfer.

15. (a) Virus Resistance in Transgenic Plants.

(*Or*)

(b) Write note on fruit ripening delaying.

Part - C (3 × 10 = 30)

Answer any **three** questions.

16. Explain in details about mitochondrial genome organization.

17. Describe cryopreservation and its applications in germplasm maintenance.

18. Explain about post translation modification.

19. Give an account on Metabolomics for plant stress response.

20. Explain Molecular cloning.

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RW-6126

501204

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Biotechnology

Lab III : MOLECULAR GENETICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Define one step growth curve.

2. Mention the importance of genetic markers.

3. Define induced mutagenesis.

4. State the difference between auxotrophic and antibiotic resistant mutants.

5. Define specialized transduction. Give an example.

6. Define conjugation with an example.

7. Give any four applications of genetic mapping.

8. What is a transposon and mention the important features of transposons.

9. Mention the important features of Chromosomal and plasmid DNA.

10. Define episomes.

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) How will you isolate single colony. Briefly explain this using an example.

(*Or*)

(b) Explain briefly about one step growth curve in Teven phage.

12. (a) What is the importance of isolating antibiotic resistant and auxotrophic mutants—Explain.

(*Or*)

(b) Explain induced mutagenesis.

13. (a) Comment briefly on specialized transducing phage.

(*Or*)

(b) Correlate specialized and generalized transduction with an example.

14. (a) State the importance of conjugation in molecular genetics.

(*Or*)

(b) Explain conjugation with an example.

15. (a) What is mutation ? Explain the types of mutation.

(*Or*)

(b) Briefly comment on transposons and their types.

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. Describe in detail about the growth rate, growth curve and titration of phages.

17. Has induced mutation has any role on the antibiotic resistant and auxotrophic mutants—Explain.

18. Explain the genetic mapping procedure.

19. Describe the mechanism of conjugation with examples.

20. Describe in detail about the transposon mutagenesis of Chromosomal and plasmid DNA.

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